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NIHVA Technologies Pvt. Ltd.




# NIHVA

# DOCK-LEVELER

## OPERATION & MAINTENANCE MANUAL

DOCK LEVELER – NDL-025,026

**CLIENT: - Jolly Containers**

| | |
|---|---|
| **DOCK LEVELER** | **MANUAL** |

## A DESPATCH & UNLOADING

1. Loading, for dispatch, will be our responsibility. On receipt of the Dock-leveler at site, please check and ensure that there is no physical damage to the equipment.

2. While unloading the equipment from the transporting vehicle, please ensure to use slings with rated capacity higher than the gross weight of the Stationary lift in packed condition.

3. Slings MUST be hooked ONLY at specified positions. These positions are indicated and marked with a figure of 'HOOK'. Improper sling position or improper slinging could cause damage to the equipment.

## B. ORDERING OF SPARES

1. When ordering for spare, please specify the following in your Purchase Order.

   - Equipment Name
   - Model Number
   - Equipment Serial No.
   - Part name / Description
   - Part Number
   - Quantity required
   - Delivery period / time
   - Mode of dispatch

   This would avoid ambiguity and help us serve you better.

## DATASHEET FOR DOCK LEVELER – NDL-025,026

**CLIENT: - Jolly Containers**

**1.0 CONSTRUCTION DETAILS**

| | |
|---|---|
| 1.1 Required Capacity - each | : 09 ton |
| 1.2 Model No. | : NDL-025,026 |
| 1.3 Leveler operation mode operated | : AC power pack hydraulically |
| 1.4 Lip operation mode | :hydraulic |
| 1.5 Operation control | : by push buttons and control panel |
| 1.6 Loading platform height from road Level | : 1200 mm (Will be confirmed during Detailing) |
| 1.7 Travel above & below dock level | : (+) 350 / (-) 250 mm |
| 1.8 Platform Size (w x l) | : 2030 x 2720 mm |
| 1.9 Lip size (w x l) | : 2030 x 400 mm |
| 1.10 Working length | : 3120 mm |
| 1.11 Overall working size | : 2030x3120 mm |
| 1.12 Maximum slope can be achieved | : 8 to 10% |
| 1.13 Clear Pit dimensions (W x L x D) | : 2080x 2780 x 485 mm |
| 1.14 Hydraulic power pack | |
| 1.14.1 Make | "NIHVA" |
| 1.14.2 Driver type | : Direct by electric Motor |
| 1.14.3 Absorbed power | : 0.75 kW |
| 1.14.4 Motor Ratings | : 0.75/1 kW /HP |
| 1.14.5 Degree of protection for motor: IP 55 | |
| 1.14.6 Class of Insulation for motor | : Class F with temp. Rise limited to Class B |
| 1.14.7 Motor RPM | : 1410 |
| 1.14.8 Pump design flow | : 1.8 LPM |
| 1.14.9 Pump pressure | : 200 (max) Kg/cm²g |
| 1.14.10 Safety valve set pressure | : 90 Kg/cm²g |
| 1.14.11 in built safety valve | : Yes |
| 1.14.12 Pump type | : Gear |
| 1.14.13 Valves required | : By vendor |
| 1.14.14 Volume of oil receiver | : 5 Lit |
| 1.14.15 Dimensions of power pack Unit | : 400 H x 560 L x 340 W mm |
| 1.14.16 First fill of oil / Qty | : By Vendor |
| 1.14.17 Hydraulic tubing | : By Vendor |
| 1.14.18 Instrumentation | : By vendor |
| 1.14.19 Control panel | : By vendor |

# DOCK LEVELLER

## INTRODUCTION:-

The hydraulic Dock Leveler is the result of many years' experience in materials handling engineering. "NIHVA " offers the best possible factor of safety with simple, easy to use controls. The high quality of manufacture gives a robust construction and many years of useful life. Each lip and platform movement can be individually controlled (or corrected) by push button.

The dock leveler can be set at any position. During loading, the Dock Leveler is in floating mode i.e. it follows the up/down movements of the vehicle deck. Even with lopsided vehicles, the Dock leveler lip will remain in contact, and toe-traps are avoided by the tilt built into the Dock Leveler.

The whole platform is designed to flex laterally adequately to adapt to sideways tilted vehicles. The leveler always seals perfectly to the lorry bed.

**PRINCIPLE:-**

Dock Leveler is an adjustable platform used between the transport vehicle and dispatch area of the warehouse of factory for efficient and safe material transfer applications. It bridges the gap between the dispatch area and the transport vehicle and compensates for variations in the truck bed height.

## CONSTRUCTION:-

It basically consists of heavy duty fabricated Top Chequered platform and bottom angle channel construction frame. Both are joined by stainless steel machined pins. Lip is hinged to the top platform by means of stainless steel pins.

Top Platform is reinforced with I Beam welded below the platform to strengthen the platform. The cylinder is mounted at the best advantage point against the top platform and bottom frame. Compact power pack includes a single HP motor with overload protection. Hydraulic Pump is Dowty make. Hydraulic operations are by using solenoid control valve.

## FEATURES:-

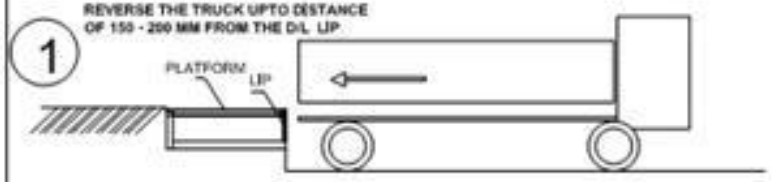
1. Convenient location of all controls.
2. Independent deck and lip operation.
3. Controls for quicker cycling time.
4. Smooth dependable hydraulics.
5. Quite operation at the touch of the button.
6. Lower operating cost and fewer maintenance requirements.
7. Recommended for container & loading unloading applications.
8. Maintenance Strut.
9. Toe guards.
10. Heavy thickness of the lip gives good life.
11. Self standing structure transfers load to the base of the pit, independent of pit skirting civil work.
12. Profiled hinges gives good welding strength.
13. No cross support imparts good flexibility.

## PRE – OPERATING INSTRUCTIONS:-

1. Ensure the dock platform is empty.
2. Ensure the truck is parked properly.
3. Ensure the chocks to the truck to arrest the movement while loading.
4. Ensure proper functioning of the switches.
5. Ensure the truck height is within the range.
6. Ensure no joint leakage.

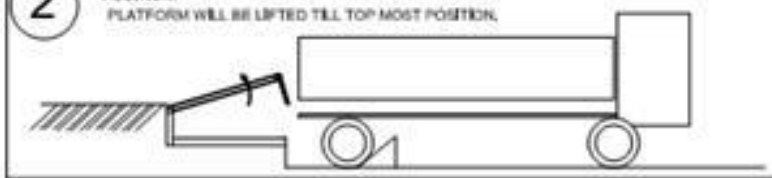
# DOCK LEVELER MANUAL

**1** REVERSE THE TRUCK UPTO DISTANCE OF 150 - 200 MM FROM THE D/L LIP




NOTE : ENSURE THAT EDGES OF TRUCK LOAD BODY ARE OUTSIDE LIP WIDTH I.E. CHECK ALIGNMENT OF LIP WITH TRUCK BODY.

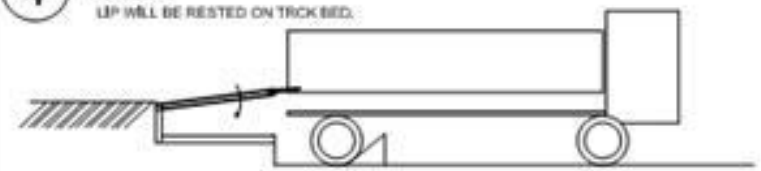
**2** HOLD PLATFORM UP BUTTON IN PRESSED POSITION.
PLATFORM WILL BE LIFTED TILL TOP MOST POSITION.



**3** HOLD LIP UP BUTTON IN PRESSED POSITION.
LIP WILL BE LIFTED TILL IT STOP.

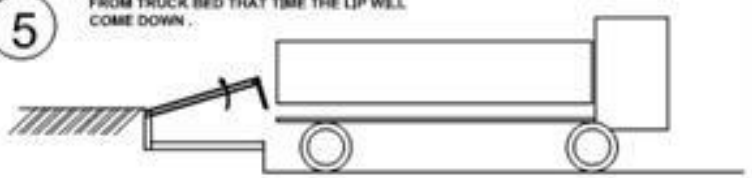


**4** HOLD PLATFORM DOWN BUTTON IN PRESSED POSITION.
LIP WILL BE RESTED ON TRCK BED.

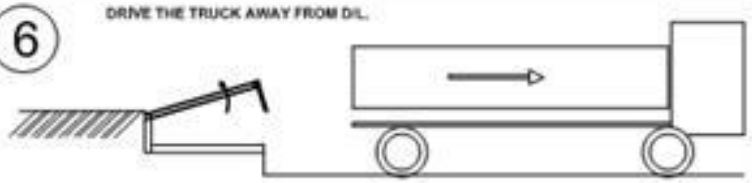


UNLOAD OR LOAD THE MATERIAL FROM THE TRUCK.

**5** PRESS PLATFORM UP TO CLEAR THE LIP FROM TRUCK BED THAT TIME THE LIP WILL COME DOWN .



**6** DRIVE THE TRUCK AWAY FROM D/L.



**7** HOLD PLATFORM DOWN BUTTON .
LIP WILL BE PARKED TO ITS ORIGINAL POSITION.



**IMPORTANT NOTICE**

**Do Not Load the Dock Platform in Raised Position without resting Lip on Truck Bed.**

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# ROUTINE MAINTENANCE AND LUBRICATION

These recommendations are applicable to the Dock Leveler considering average daily usage of 15 to 20 times. If it is used more frequently, the same time intervals for the maintenance should be adjusted accordingly.

**DAILY:-**

1. Check Hydraulic pipe lines for leak proof working.
2. Check electrical connections for their proper working.

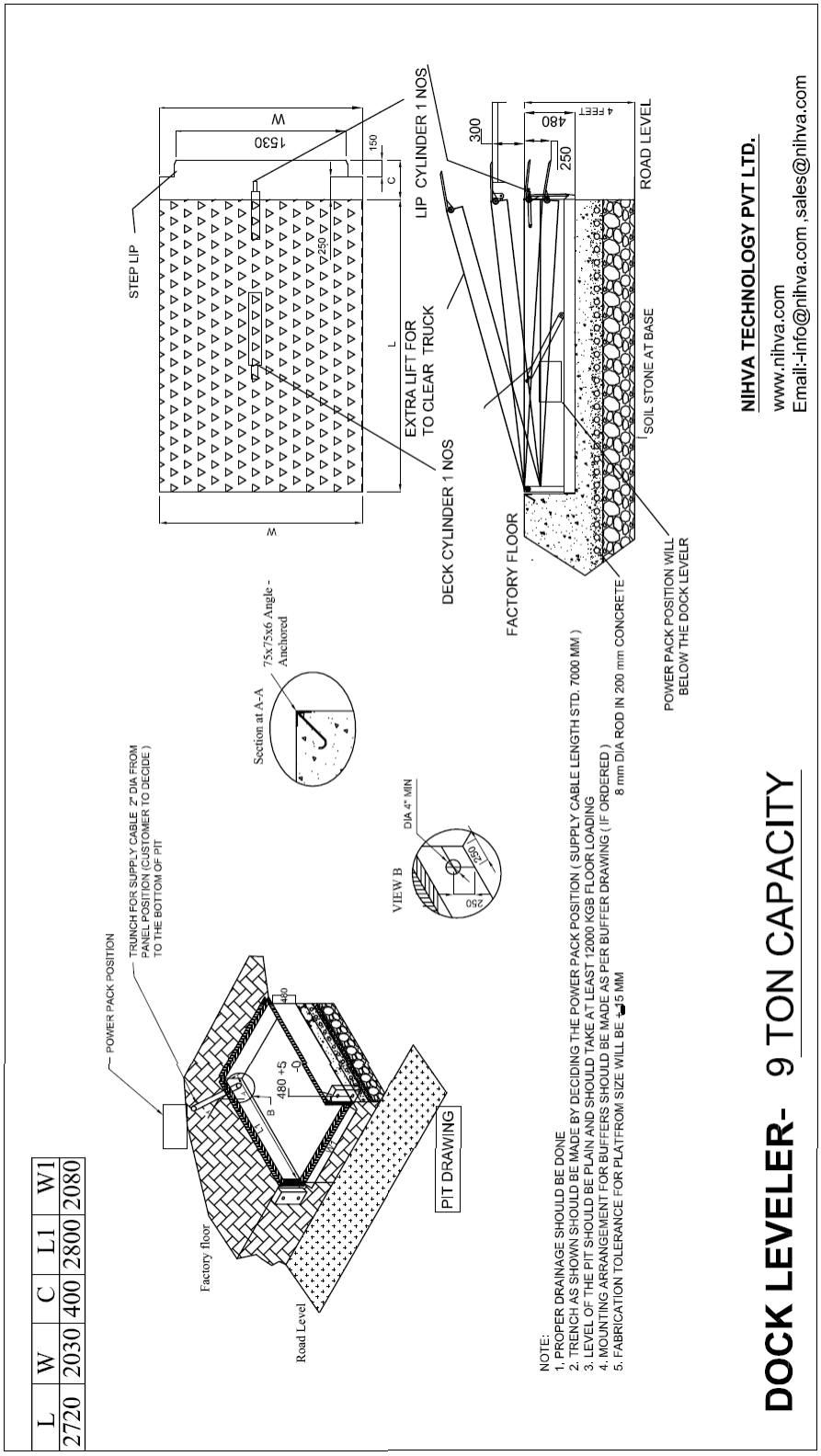
**MONTHLY:-**

1. Grease the hinge pins at bottom frame and top plate joint and extension lip plate hinge joint.
2. Grease the main Jack Cylinder and Lip Jack Cylinder joints.

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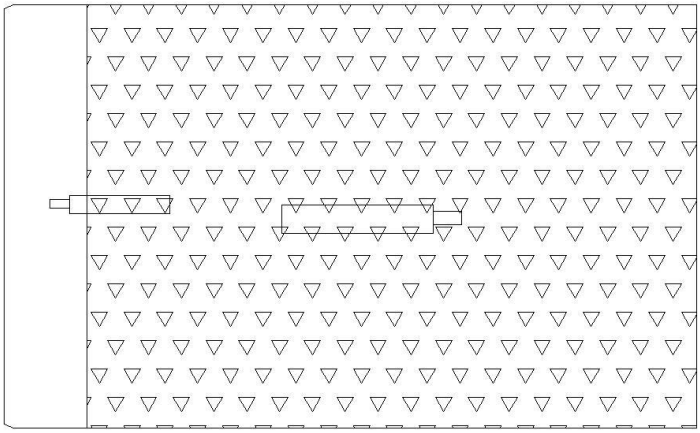
# ASSEMBLIES & SPARE PARTS LIST

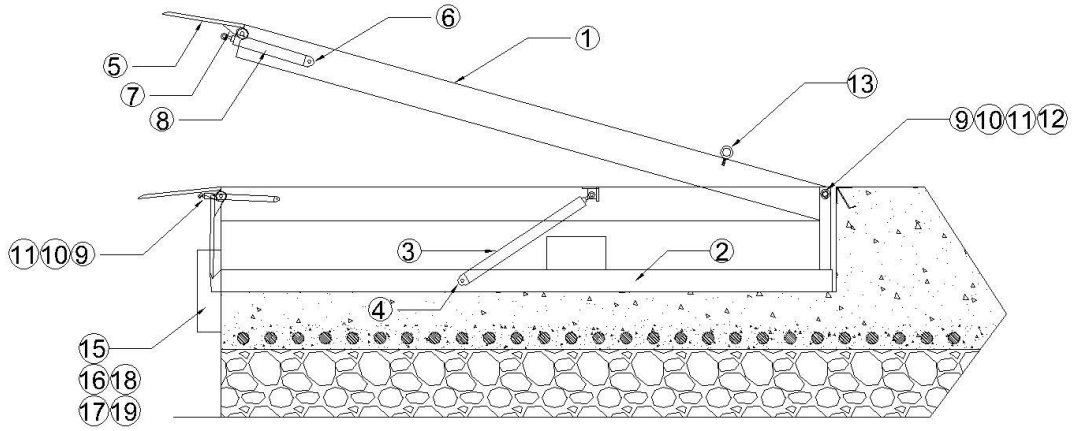
| DOCK LEVELER | MANUAL |
|---|---|




| L | W | C | L1 | W1 |
|---|---|---|---|---|
| 2720 | 2030 | 400 | 2800 | 2080 |

# DOCK LEVELER | MANUAL



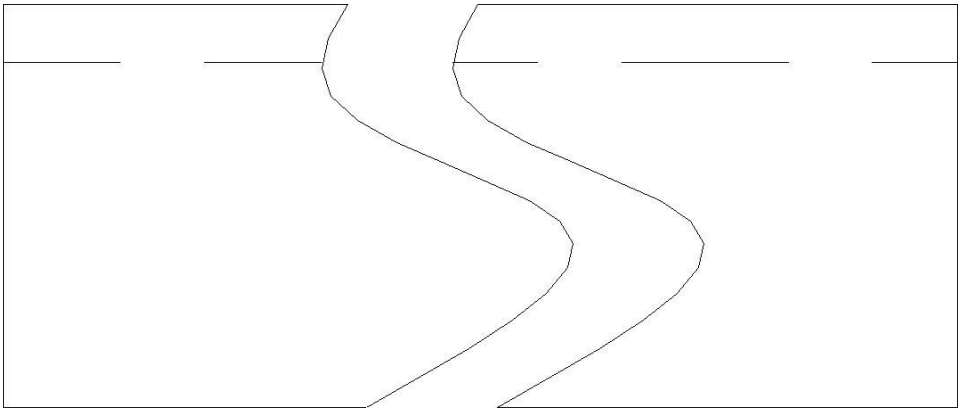


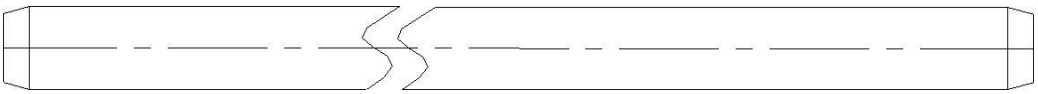

# DOCK LEVELER

# MANUAL

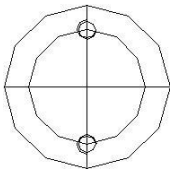
| SR. NO. | DESCRIPTION | QTY. |
|---|---|---|
| 1 | TOP PLAT FORM WELDED ASSLY | 1 |
| 2 | BOTTOM FRAME WELDED ASSLY | 1 |
| 3 | MAIN JACK ASSLY | 2 |
| 4 | MAIN JACK PIVOT PIN | 4 |
| 5 | LIP PLATE | 1 |
| 6 | LIP JACK BOTTOM PIVOT PI N | 1 |
| 7 | LIP JACK TOP PIVOT PIN | 1 |
| 8 | LIP JACK ASSLY | 1 |
| 9 | PLATFORM & LIP PIVOT PIN | 2 |
| 10 | PIN LOCKING WASHER | 2 |
| 11 | M6 X 20 L CSK BOLT | 8 |
| 12 | GREASE NIPPLE | 6 |
| 13 | I BOLT M16 | 4 |
| 14 | ANCHOR FASTNER | 8 |
| 15 | RUBBER BUFFER | 2 |
| 16 | RUBBER BUFFER PIVOT PIN | 4 |
| 17 | SPLIT PIN | 8 |
| 18 | RUBBER BUFFER MTG. ANGLE | 4 |
| 19 | ANCHOR FASTNER | 8 |

SPLIT LIP PLATE
PART NO-70000160
QTY-01

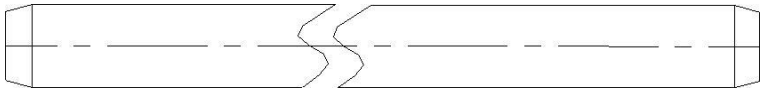




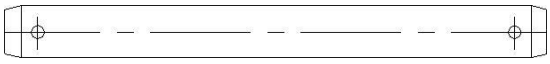
PLATFORM PIVOT PIN (REAR)
PART NO-70000161
QTY-01



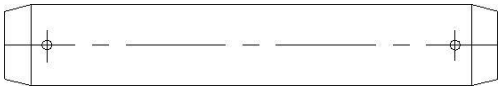
PIN LOCKING WASHER
PART NO-70000164
QTY-04



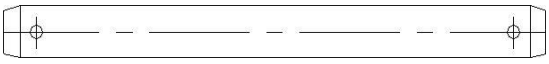
LIP PLATE PIVOT PIN (M.S)
PART NO-70000162
QTY-01



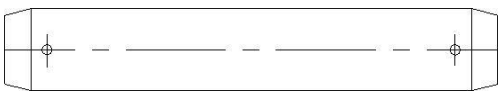
LIP JACK PIVOT PIN( TOP)
PART NO-70000167 QTY-
01



MAIN JACK PIVOT PIN( TOP)
PART NO-70000165
QTY-01

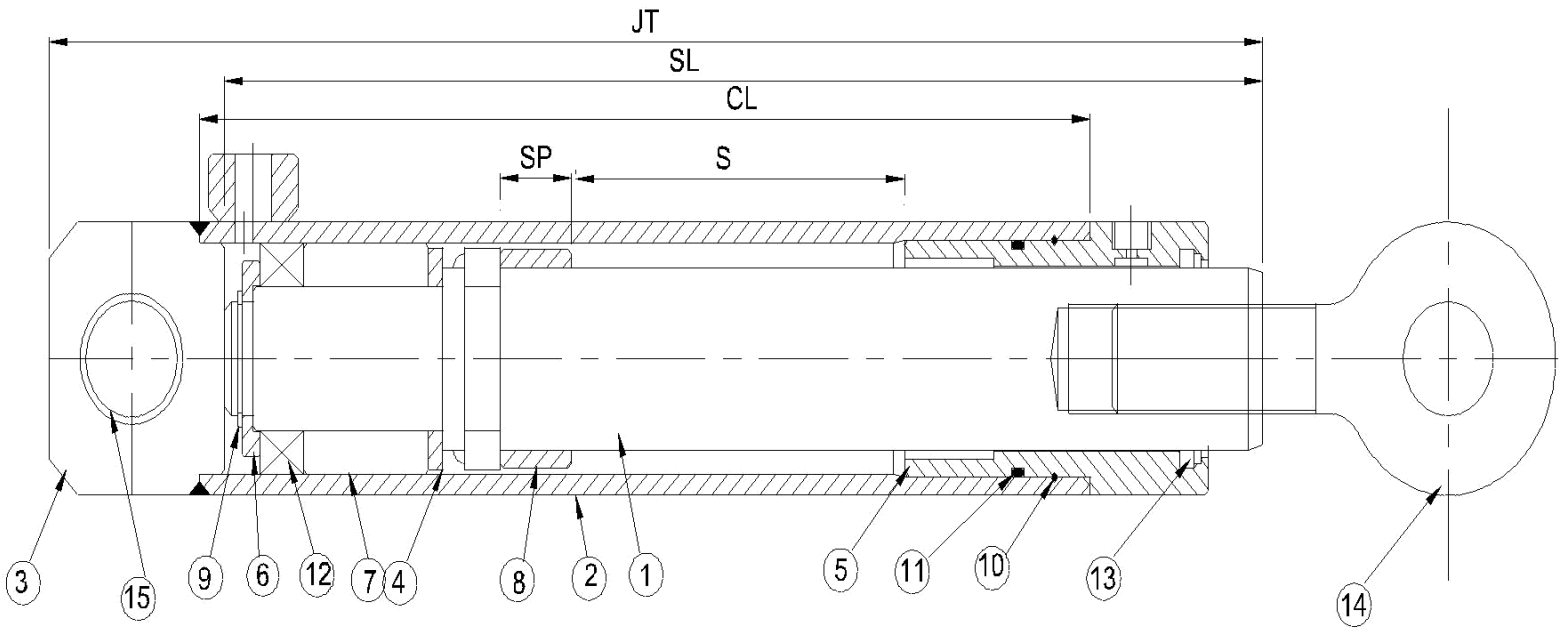


LIP JACK PIVOT PIN(BOTTOM )
PART NO-70000168
QTY-01



MAIN JACK PIVOT PIN(BOTTOM )
PART NO-70000166
QTY-01

# DOCK LEVELER MANUAL



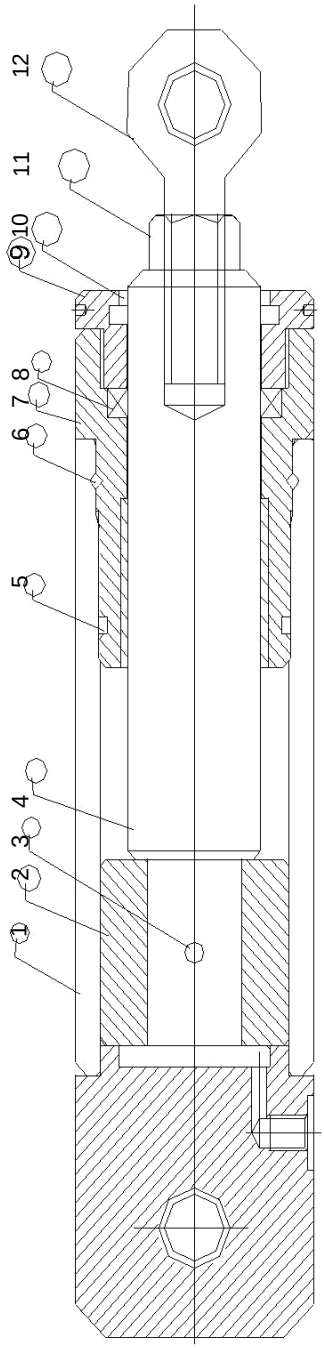

# MAIN JACK ASSLY

## PART NO-70000156

| SR.NO. | ITEM NO. | PARTS NO. | DESCRIPTION | SIZE | QTY |
|---|---|---|---|---|---|
| 15 | 15 | 10000071 | SINTERED BUSH | 1"X1.1/8"X1" | 2 |
| 14 | 14 | 10000070 | ROD END BEARING | M24X2XPH25PIN | 1 |
| 13 | 13 | WPS00015 | WIPER SEAL | Ø40xØ48x7 | 1 |
| 12 | 12 | PTS00011 | PISTON U SEAL | Ø50xØ40x9 | 1 |
| 11 | 11 | ORN00040 | O RING | Ø48X2.5 | 1 |
| 10 | 10 | LKW13225 | LOCKING WIRE | 13 GAUGE | 1 |
| 9 | 9 | CP00009 | CIRCLIP | EXT Ø25 | 1 |
| 8 | 8 | 70000050 | SPACER Ø40 | Ø40XØ49XSP | 1 |
| 7 | 7 | 70000154 | PISTON Ø40 | Ø50XØ40X52 | 1 |
| 6 | 6 | 70000049 | WAHSER RETAINER SEAL | Ø48XØ25.2X5 | 1 |
| 5 | 5 | 70000048 | WIPER SEAL HOUSING | Ø63X82 L | 1 |
| 4 | 4 | 70000047 | WASHER | Ø49XØ32X4 | 1 |
| 3 | 3 | 70000159 | BOTTOM END | Ø63X49 | 1 |
| 2 | 2 | 70000158 | ASSLY WELDED CYLINDER | 2" HONEDx | 1 |
| 1 | 1 | 70000157 | SHAFT JACK | Ø40 x 580 | 1 |

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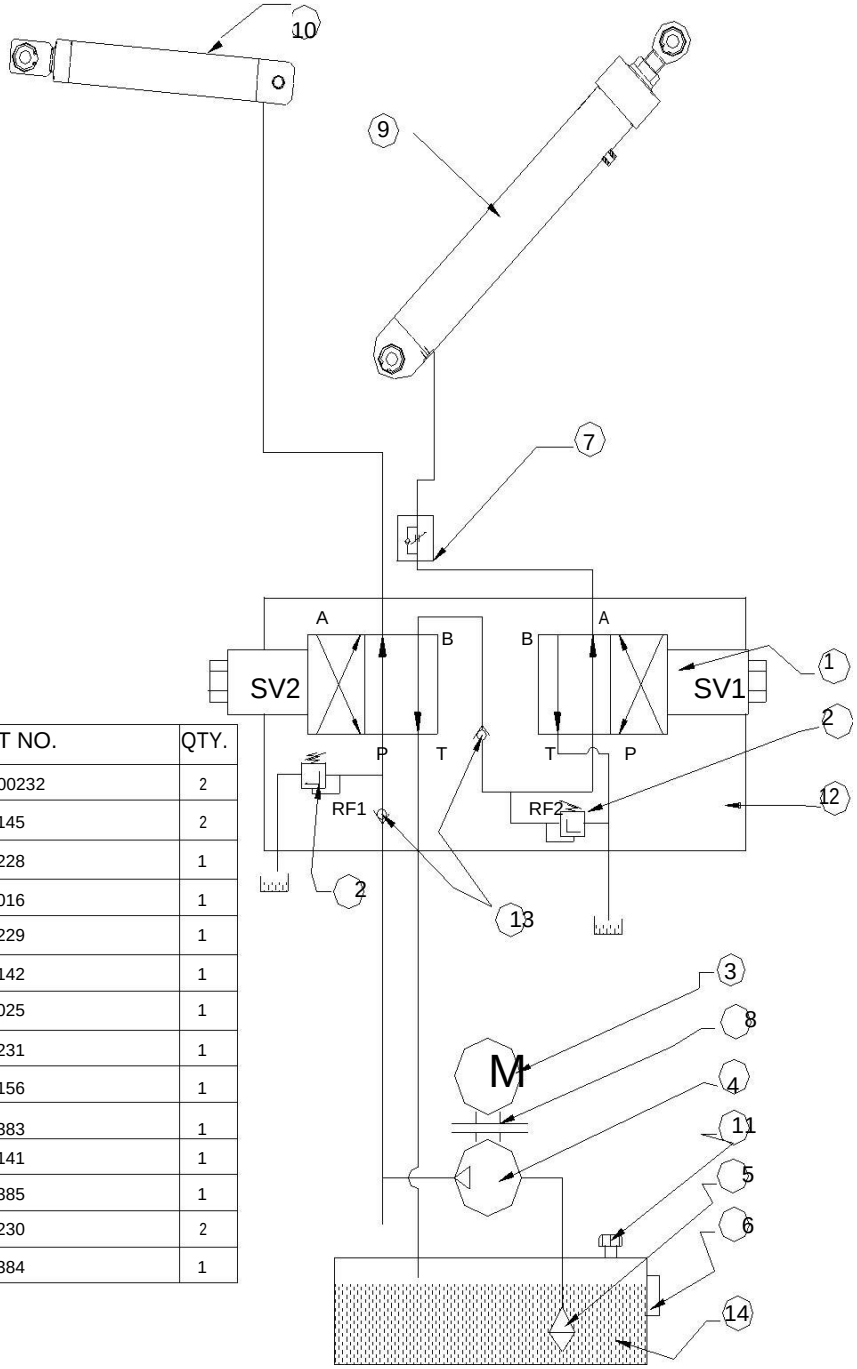

| SR.NO. | PARTS NO. | DESCRIPTION | QTY |
|---|---|---|---|
| 12 | REB00006 | ROD END BEARING | 1 |
| 11 | | HEX NUT M-16X2P | 1 |
| 10 | WPS00002 | SEAL WIPER | 1 |
| 9 | 70000390 | WIPER SEAL HOUSING | 1 |
| 8 | RDS00003 | SEAL ROD | 1 |
| 7 | 70000389 | SEAL HOUSING | 1 |
| 6 | LKW13235 | WIRE LOCKING | 1 |
| 5 | ORN00002 | O RING | 1 |
| 4 | 70000388 | SHAFT(PLATED) | 1 |
| 3 | DSP00001 | SPRING PIN | 1 |
| 2 | 70000387 | PISTON | 1 |
| 1 | 70000386 | JACK CYLINDER | 1 |

# LIP JACK ASSLY

PART NO-70000383

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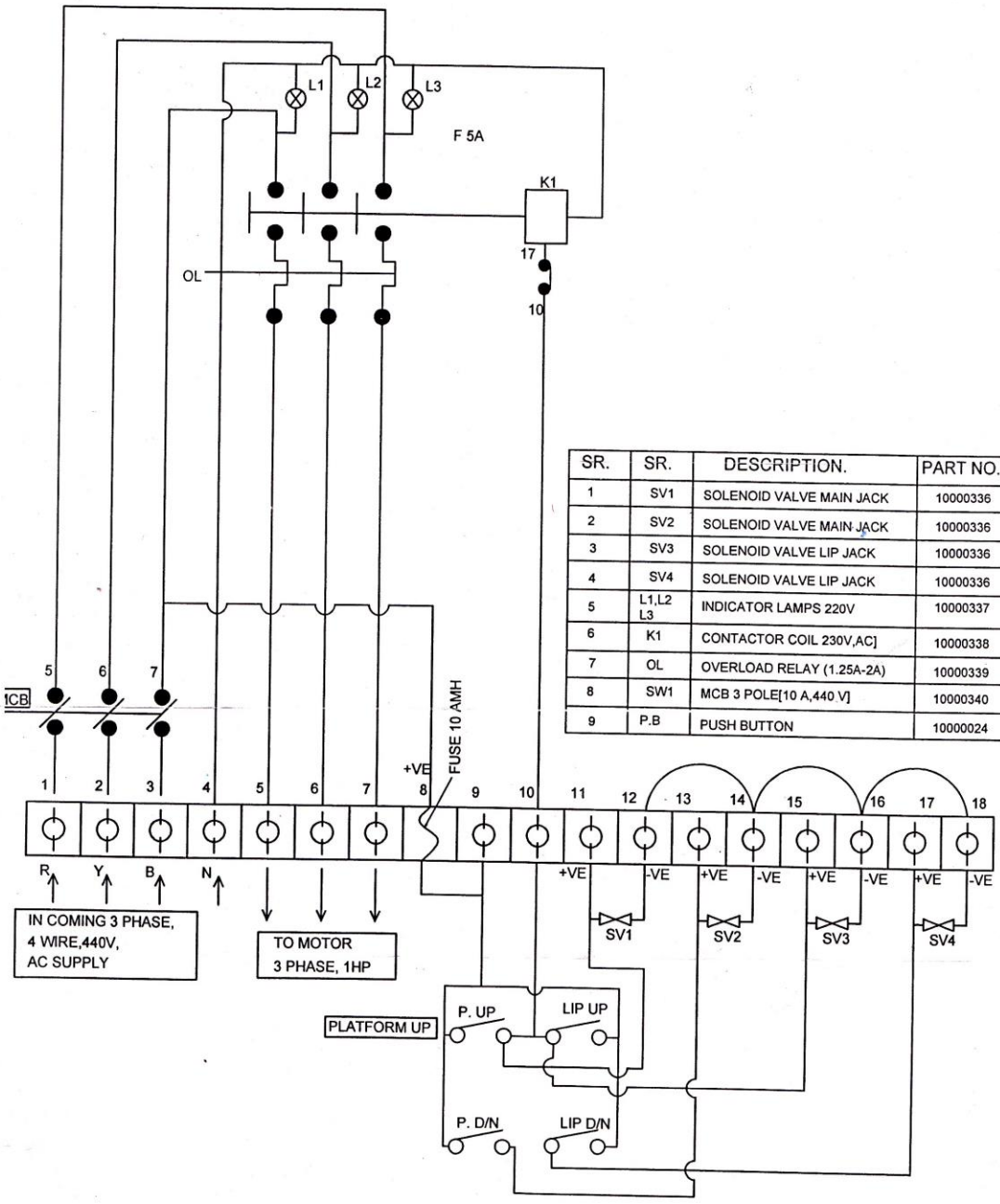
# HYDRAULIC CIRCUIT FOR DOCK LEVELER




| SR.NO. | DESCRIPTION. | PART NO. | QTY. |
|---|---|---|---|
| 1 | SOLENOID VALVE (SV1,SV2) | 10000232 | 2 |
| 2 | RELIEF VALVE (RF1,RF2) | 10000145 | 2 |
| 3 | MOTOR 1HP,3PHASE | 10000228 | 1 |
| 4 | GEAR PUMP(OP-3006) | 10000016 | 1 |
| 5 | SUCTION STRANER | 10000229 | 1 |
| 6 | OIL LEVEL INDICATOR | 10000142 | 1 |
| 7 | THROTTLE VALVE | 10000025 | 1 |
| 8 | LOVEJOY COUPLING | 10000231 | 1 |
| 9 | MAIN JACK ASSLY. | 70000156 | 1 |
| 10 | LIP JACK ASSLY. | 70000383 | 1 |
| 11 | BREATHER FSB-25 | 10000141 | 1 |
| 12 | MANIFOLD BLOCK | 70000385 | 1 |
| 13 | IN LINE CHECK VALVE 1/4" | 10000230 | 2 |
| 14 | OIL TANK 15LTR CAPACITY | 70000384 | 1 |

**DOCK LEVELER** | **MANUAL**

## DOCK LEVLER 4SOLENOID VALVE AC PANEL




| SR. | SR. | DESCRIPTION. | PART NO. | QTY. |
|---|---|---|---|---|
| 1 | SV1 | SOLENOID VALVE MAIN JACK | 10000336 | 1 |
| 2 | SV2 | SOLENOID VALVE MAIN JACK | 10000336 | 1 |
| 3 | SV3 | SOLENOID VALVE LIP JACK | 10000336 | 1 |
| 4 | SV4 | SOLENOID VALVE LIP JACK | 10000336 | 1 |
| 5 | L1,L2 L3 | INDICATOR LAMPS 220V | 10000337 | 3 |
| 6 | K1 | CONTACTOR COIL 230V,AC] | 10000338 | 1 |
| 7 | OL | OVERLOAD RELAY (1.25A-2A) | 10000339 | 1 |
| 8 | SW1 | MCB 3 POLE[10 A,440 V] | 10000340 | 1 |
| 9 | P.B | PUSH BUTTON | 10000024 | 1 |

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